

PRACTICAL-9

AIM:- To study about the configuration of link state EIGRP routing protocol by using Cisco Packet Tracer.

● **STEPS :**

1. Assign IP Addresses to PCs

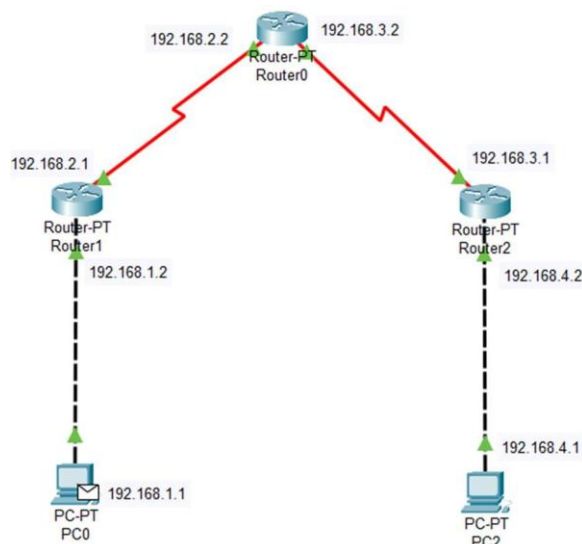
Each PC must have an IP address in the correct subnet and a default gateway (the router it connects to).

On PC0 (Connected to Router1)

1. Go to **PC0** in Packet Tracer.
2. Open **Desktop > IP Configuration**.
3. Assign the following:
 - **IP Address:** 192.168.1.1
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.1.2 (Router1)

On PC1 (Connected to Router2)

1. Go to **PC1** in Packet Tracer.
2. Open **Desktop > IP Configuration**.
3. Assign the following:
 - **IP Address:** 192.168.4.1
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.4.2 (Router2)



2. Assign IP Addresses to Router Interfaces

Each router has multiple interfaces connected to different subnets. Configure them correctly.

On Router1

1. Enter CLI mode and configure interfaces:

```
enable
configure terminal
interface FastEthernet0/0
ip address 192.168.1.2 255.255.255.0
no shutdown
exit
interface Serial0/0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

On Router0

```
enable
configure terminal
interface Serial0/0/0
ip address 192.168.2.2 255.255.255.0
no shutdown
exit
interface Serial0/0/1
ip address 192.168.3.2 255.255.255.0
no shutdown
exit
```

On Router2

```
enable
configure terminal
interface FastEthernet0/0
ip address 192.168.4.2 255.255.255.0
no shutdown
exit
interface Serial0/0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
```

3. Configure EIGRP on Each Router

Enable **EIGRP** and advertise the directly connected networks.

On Router1

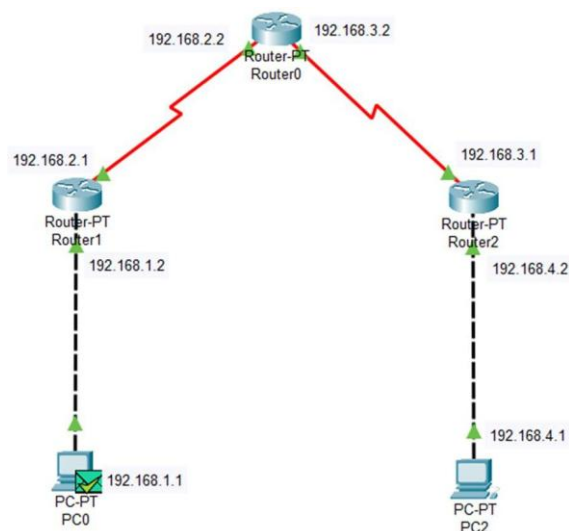
```
enable
configure terminal
router eigrp 100
network 192.168.1.0 0.0.0.255
network 192.168.2.0 0.0.0.255
no auto-summary
exit
```

On Router0

```
enable
configure terminal
router eigrp 100
network 192.168.2.0 0.0.0.255
network 192.168.3.0 0.0.0.255
no auto-summary
exit
```

On Router2

```
enable
configure terminal
router eigrp 100
network 192.168.3.0 0.0.0.255
network 192.168.4.0 0.0.0.255
no auto-summary
exit
```



4. Verify Configuration

Check Neighbor Relationships

On each router, run:

```
show ip eigrp neighbors
```

This should display the adjacent routers.

Check Routing Table

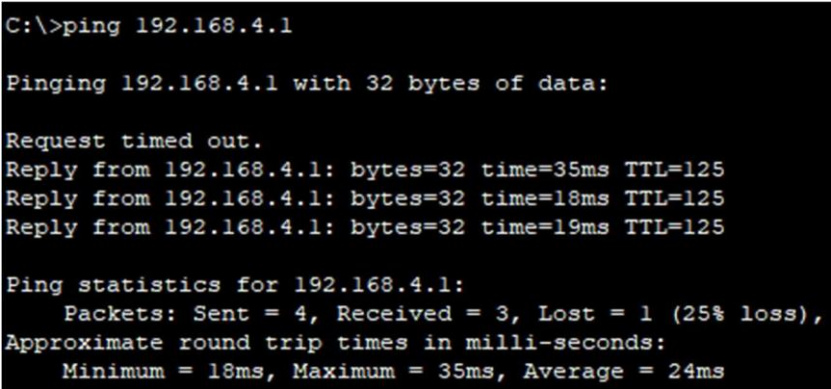
```
show ip route eigrp
```

It should show routes learned via **EIGRP**.

Test Connectivity

1. From PC0 (192.168.1.1), ping PC1 (192.168.4.1):

```
ping 192.168.4.1
```



```
C:\>ping 192.168.4.1

Pinging 192.168.4.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.1: bytes=32 time=35ms TTL=125
Reply from 192.168.4.1: bytes=32 time=18ms TTL=125
Reply from 192.168.4.1: bytes=32 time=19ms TTL=125

Ping statistics for 192.168.4.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 18ms, Maximum = 35ms, Average = 24ms
```